

INTRODUCTION TO ENGINEERING • LECTURE 1

Engineering as a Profession

Culture, Identity, and the Historical, Social, and Economic Role of Engineering

Dr. Mohammad Reza Bahrami

Dean, Faculty of Energy Engineering — Tashkent University of Technology

Adjunct Professor, Constructor University Bremen

Fall 2026

Disclaimer: this lecture references material adapted from Kosky, Balmer, Keat & Wise, *Exploring Engineering* (5th ed.), and from the instructor's own course slides, with modifications. No authorship of third-party material is claimed.

About Me

- **Professor**
Innopolis University
- **Dean, Faculty of Engineering**
Tashkent University of Technology (TashTech)
- **Adjunct Professor**
Constructor University, Bremen, Germany
- **PhD in Mechanical Engineering**
Currently completing a Doctor of Science (DSc)
- **Research interests**
Optimal control, autonomous systems, computational geometry, model predictive control, reinforcement learning
- **Active researcher**
Publications include IEEE Access and other peer-reviewed venues



Dr. Mohammad Reza Bahrami

Professor, Dean of Energy Engineering

My Path Into Engineering



- **Site:** sites.google.com/view/mohammadrezabahrami
- **ResearchGate:** researchgate.net/profile/Mohammad-Reza-Bahrami
- **Google Scholar:** scholar.google.com (search “Mohammad Reza Bahrami”)
- **Reach out any time with questions about the course or engineering as a career.**

Agenda



Why Engineering?

An interactive warm-up



Historical Role

From ancient builders to the digital age



Economic Role

Scale, salaries, and national development



Science, Engineering & Design

How the three connect — and differ



What Is Engineering?

Etymology, definitions, and fields



Social Role

Engineering, society, and the Grand Challenges



Culture, Identity & Profession

Traits, societies, and a code of ethics



Wrap-Up & Preview

Key takeaways and what's next

Why Do You Want to Be an Engineer?

- Open the form on your phone or laptop.
- Answer in one or two words — be honest, this isn't graded.
- We'll read a few responses out loud before moving on.

Think about it:

Was it a person who inspired you? A problem you wanted to solve? A subject you were good at in school?

Time

50%

Live poll — Google Form

<https://forms.gle/cCnAZNknys1VKA6u8>



Where the Word “Engineer” Comes From



ingenium

(Latin) — “the product of genius”

ingeniare

(Latin) — “to contrive, to devise”

→ engine → “engine-er”

Same root today: *ingénieur* (FR/DE) · *ingegnere* (IT)

- About 2,000 years ago, *ingenium* described the design of a new machine — a product of cleverness.
- The word soon broadened to describe machines themselves, then the people who designed them: “engine-ers.”
- **So at its root, the word “engineer” means:**

“A creative, ingenious person who finds solutions to practical problems.”

Defining Engineering Today

Engineering is the practice of using natural science, mathematics, and the engineering design process to solve technical problems, increase efficiency and productivity, and improve systems.

“Science can amuse and fascinate us all, but it is engineering that changes the world.”

— Isaac Asimov

“Scientists investigate that which already is; engineers create that which has never been.”

— Albert Einstein

WHAT IS ENGINEERING?

One Word, Many Fields

Because engineering touches everything around us, the discipline branches into dozens of specialized fields.



Civil

Roads, bridges, water systems



Mechanical

Machines, engines, robots



Electrical

Power, circuits, controls



Computer

Hardware & software systems



Chemical

Processes, materials, fuels



Aerospace

Aircraft & spacecraft



Biomedical

Medical devices & health tech



Environmental

Water, air, sustainability

...plus agricultural, ceramic, industrial, manufacturing, marine, materials, mining, nuclear, petroleum, textile, and more.

A Common Misconception



NOT THE SAME AS

A plumber installs and repairs pipes.

ENGINEERING

An engineer designs the plumbing system — sizing pipes, pressures, and layouts before anything is built.



NOT THE SAME AS

An auto mechanic repairs a car.

ENGINEERING

An engineer designs the parts — the engine, transmission, and safety systems the mechanic later services.

Which Field Interests You Most?

- Vote for one field from the list on the previous slide (or write in your own).
- We'll look at the live results together as a class.
- No wrong answers — this just helps us get to know the room.

50%

Time





PART 2

The Historical Role of Engineering

From the first walls and aqueducts to the machines that built the modern world.

Engineering's Ancient Roots

- The oldest form of engineering is military engineering — fortifications, siege works, and roads built for armies.
- The first non-military engineers were called civilian engineers — later shortened to “civil engineers.”
- Long before the word existed, the practice did: the pyramids, aqueducts, and city walls of antiquity were engineering achievements.
- **Civil engineering remains one of the oldest engineering disciplines in continuous practice today.**

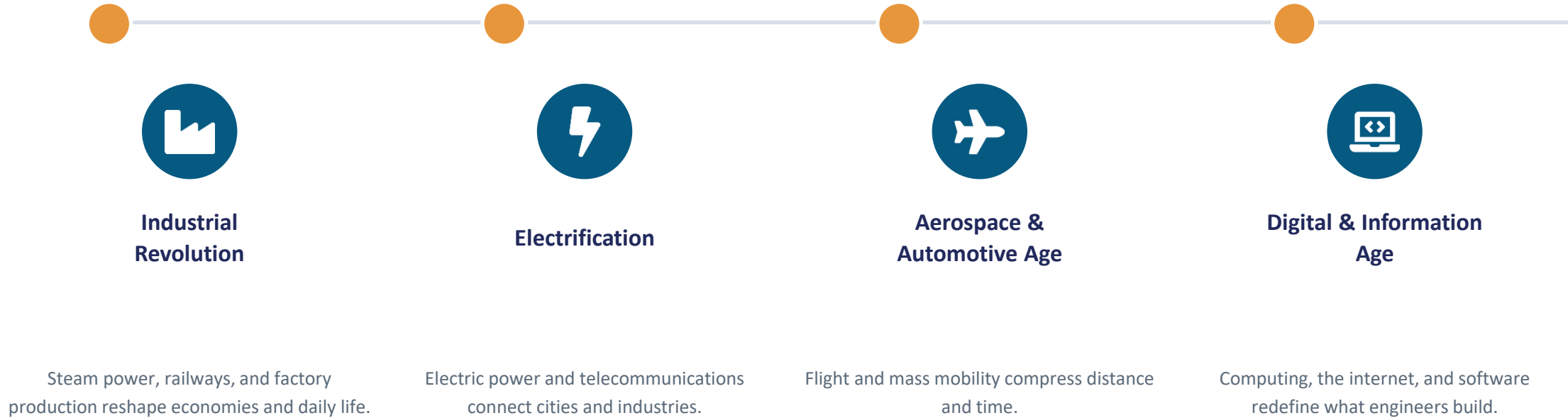


Discuss

Name one ancient or historical structure you consider a great feat of engineering. What problem did it solve?

[2–3 minutes — a few volunteers share out loud]

From Steam to Silicon



In every era, engineers were the ones who converted new scientific knowledge into tools, systems, and infrastructure people could actually use.



“The engineer has been, and is, a maker of history.”

— James Kip Finch, engineer and educator

Every bridge, power grid, and communications network reflects a decision an engineer made — decisions that quietly shaped how civilizations grew, traded, and connected.



PART 3

The Social Role of Engineering

Engineers sit at the interface between technical possibility and human need.

Engineering & Society

- Engineering shapes the everyday conditions of human life: the water we drink, the power in our homes, the hospitals that treat us, and the roads that connect us.
- The U.S. National Academy of Engineering's “Grand Challenges” frame engineering's social mission for this century:



Provide access to clean water



Restore & improve urban infrastructure



Advance health informatics



Make solar energy economical



Secure cyberspace



Engineer better medicines

(6 of 14 Grand Challenges — the full list also covers energy, the brain, learning, and scientific discovery.)

The Interface Between Science and Society

“Engineers operate at the interface between science and society.”

— Gordon Stanley Brown, MIT engineering professor

- Because engineers translate scientific possibility into products, systems, and infrastructure that reach millions of people, their decisions carry direct social consequences — for safety, health, equity, and the environment.
- That is precisely why engineering is organized as a profession, with codes of ethics and standards of practice — a topic we'll return to shortly.



PART 4

The Economic Role of Engineering

One of the largest, best-compensated, and most consequential professions in the modern economy.

Engineering's Economic Footprint

1.6M+

engineers working in
the United States alone

#2

largest profession
in the U.S. by size

\$68K

avg. starting salary for
U.S. engineering grads (2017)

- **An engineering degree is also a launchpad: engineering graduates regularly move into medicine, law, business, and public office, carrying their problem-solving training with them.**
- **Beyond individual careers, engineering drives national development — it is how countries build the infrastructure, industry, and technology base that economic growth depends on.**

One Way Engineering Touched Your Day

- Think of one ordinary moment from today — getting water, charging your phone, riding the bus.
- Write one sentence describing the engineering behind it.
- We'll compile answers into a shared list for the class.

50%

Time

<https://forms.gle/BSa4eT5gcmDdVkFj6>





PART 5

Engineering Culture, Identity & Profession

What it means to think, work, and be accountable as an engineer.

A Global Professional Culture

- The word for “engineer” is remarkably consistent across languages — ingénieur, ingegnere — reflecting a shared identity that crosses borders.
- Engineers usually work in teams, from two or three people to hundreds on a single large project, such as designing a commercial aircraft.
- Professional societies reinforce that shared identity, defining standards, continuing education, and ethics for each discipline:

ASME

Mechanical

IEEE

Electrical / Computer

ASCE

Civil

AIChE

Chemical

AIAA

Aerospace

ANS

Nuclear

The Mindset of a Good Problem Solver

Ten traits, adapted from the “DNA of Problem Solvers,” describe the shared identity of engineers across every field.



Objective



Analytical



Creative



Dedicated



Curious



Courageous



Adventurous



Enthusiastic



Patient



Vigilant

Which Trait Is Your Engineering Superpower?

- Look back at the ten traits — pick the one that feels most like you today.
- Submit it in the form; we'll see which traits are most common in the room.
- Keep this in mind: your project teams will need a mix of these strengths, not ten of the same one.

Why it matters:

Starting next week, you'll form the team you'll work with all semester on one real system. Knowing your own strengths — and your teammates' — makes that team stronger from day one.

Time

50%

<https://forms.gle/qJaifBvQDKSXSbwf8>



Engineering as a Profession

- A code of ethics lets a profession serve everyone it touches — the public, the employer, and the engineer's own integrity — even under competitive pressure to cut corners.
- **Under the NSPE Code, engineers shall:**
 - Hold paramount the safety, health, and welfare of the public
 - Perform services only in areas of their competence
 - Issue public statements only in an objective, truthful manner
 - Act as faithful agents or trustees for each client
 - Avoid deceptive acts
 - Conduct themselves honorably, to enhance the profession



Discuss

Your roommate skipped the homework and wants to copy yours before class. What do you do — and which canon of the code applies?

[Quick show of hands, then 2–3 minutes of discussion]



PART 6

Science, Engineering & Design

Three closely related ways of knowing — each with a different aim.

How Engineering Differs From Science



Science

Investigates what already exists.

Asks: why does this happen?

Seeks explanation, generalizable laws, and truth about the natural world.



Engineering

Creates what has never existed.

Asks: how can we make this work?

Applies scientific knowledge, under real constraints, to build a solution.

“Scientists investigate that which already is; engineers create that which has never been.” — Albert Einstein

How Engineering Differs From Design

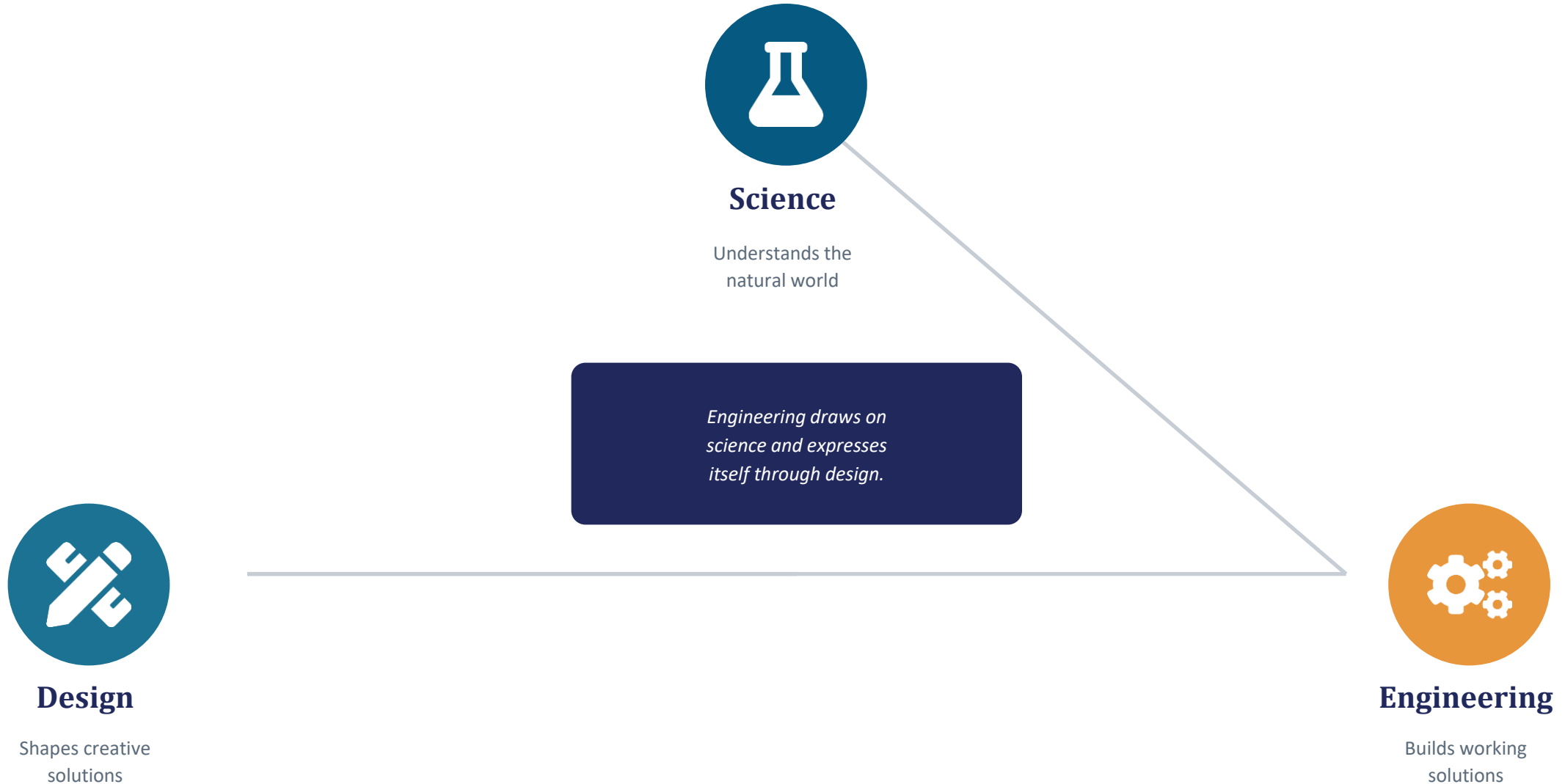
The bridge test: If a bridge is shaped to be visually stunning with no thought to whether it can carry a load — that's design without engineering. If the shape also comes from analyzing loads, materials, and structural integrity — that's engineering design.

- **Design**
- The creative, open-ended act of shaping a solution — form, aesthetics, and user experience all matter here.
- **Engineering design**
- Design disciplined by analysis: physics, materials, cost, manufacturability, and safety all constrain the possibilities.

“Design is not how it looks like and feels like. Design is how it works.”

— Steve Jobs

How the Three Connect



Scientist vs. Engineer — In Your Own Words

- In one sentence, describe the difference between a scientist and an engineer — no textbook definitions, your own words.
- We'll compare a few answers against Einstein's line from earlier.

<https://forms.gle/Zanb8ffFDNQuxZHv9>



Time

50%

Key Takeaways



“Engineer” comes from Latin roots meaning cleverness and invention — the identity is baked into the word.



Engineering has shaped every historical era, from ancient fortifications to the digital age.



Engineers sit at the interface between science and society, with direct social responsibility.



Engineering is a major economic force — one of the largest, best-paid professions worldwide.



Shared traits, societies, and a code of ethics give engineering a global professional culture.



Science explains what is; design shapes possibilities; engineering builds what works.

Independent Study — Individual Reflection



Due before Lecture 2

Identify three everyday systems around you, and for each one describe:

- **Its boundary** — what's inside the system, and what's outside it
- **Its stakeholders** — who uses it, runs it, or is affected by it

Examples to spark ideas: your dormitory's laundry room, the university bus route, a campus dining hall, a mobile payment app.

Why: this reflection feeds directly into next week's team task — choosing the one system your team may study for the rest of the semester.

COMING UP NEXT

Lecture 2: Systems Thinking I

We'll treat everyday things around you as systems — interacting elements, processes, and stakeholders — and start defining the system your team will study all semester.

Thank You — Questions?

Dr. Mohammad Reza Bahrami

Dean, Faculty of Energy Engineering — Tashkent University of Technology
sites.google.com/view/mohammadrezabahrami

What you have learn today?
A small competition: